Mapping the Machine

A Compensation, Tenure, and Structure Analysis of the District's Public Workforce

Sule Lubega · MATH 014 — Introduction to Data Science · Spring 2026 · Howard University

1. Objective

This project asks three connected questions about the District of Columbia's public workforce: (1) how does compensation vary across DC's roughly ninety agencies, and which agencies offer the highest median pay; (2) does employee tenure predict pay, and does that relationship differ across appointment categories such as Career Service, Educational Service, and Management Supervisory Service; and (3) where is internal pay dispersion most extreme inside large agencies. The objective is not to produce a single ranking but to map the structural shape of public employment in DC, to make a workforce of more than thirty-five thousand people legible through a small set of disciplined visualizations and summary statistics.

2. Introduction

The DC government employs more than forty thousand people across roughly ninety agencies, a workforce comparable in size to many Fortune 500 companies and one whose payroll is funded entirely by public dollars. The DC Public Employee Salary dataset, published openly by DC Open Data, makes that payroll inspectable: every full-time employee, their job title, their agency, their pay grade, their hire date, and their annual compensation rate. The dataset is analytically interesting for three reasons. First, it carries the kinds of real-world messiness this course has emphasized, mixed casing in string columns, sentinel values in compensation, hire dates stored as strings, and a few dozen missing grade entries. Second, it contains a clean cross-section of categorical, ordinal, continuous, and temporal variables, so it rewards both univariate and multivariate exploration. Third, the data is local: as Howard students living and working in the District, the agencies in this dataset are the agencies that run the city around us.

3. Method

3.1 Data Understanding

The raw dataset contains 40,611 rows and nine columns: the employee's first and last names, job title, agency code (DESCRSHORT), grade, compensation rate, hire-date string, appointment type, and a unique object identifier. There are no exact duplicate rows. Forty rows are missing a grade value. Compensation ranges from $0.03 (clearly a data-entry artifact) to $390,232. The hire-date column spans from 1985 to 2025 and stores dates as strings in YYYY/MM/DD format. The appointment-type column packs two pieces of information together, a category (CS, Ed, MSS, ExS, LS, Exe, Acting) and a status (Reg, Term, Temp), which we decompose during cleaning.

3.2 Data Cleaning

Cleaning proceeded in seven steps. (a) String columns were stripped of whitespace. (b) The JOBTITLE column, which arrived in mixed casing, was standardized to title case for display consistency without loss of information. (c) HIREDATE_STRING was parsed into a real datetime (HIRE_DATE) using the format string %Y/%m/%d, with errors='coerce' to surface any unparseable rows; none required dropping. (d) A salary floor of $10,000 was applied to remove the 4,984 rows with implausibly low compensation values, the lowest of which were data-entry artifacts under one dollar. The floor preserves part-time and seasonal positions while removing records that would distort group medians. (e) The forty missing GRADE values were filled with the sentinel "UNK" rather than dropped, because those rows still carry information about pay, agency, and tenure. (f) Three new features were engineered: TENURE_YEARS (years between HIRE_DATE and the analysis date of April 26, 2026), HIRE_YEAR, and HIRE_DECADE; and the appointment type was decomposed into APPT_CATEGORY and APPT_STATUS. (g) The OBJECTID and the now-redundant HIREDATE_STRING columns were dropped. The final cleaned dataset contains 35,627 rows and twelve columns and is saved as sule_final_project.csv.

4. Storytelling: Data Visualization and Interpretation

Eight visualizations underpin the analysis in the accompanying Jupyter notebook: a histogram of compensation, a bar chart of agencies by headcount, a box plot of pay by appointment category, a scatter plot of tenure versus pay, a heatmap of medians across agency × appointment, a faceted KDE plot of pay densities, an interactive Plotly treemap, and a four-dimensional bubble chart. This report reproduces five of the most informative.

4.1 The Shape of Public-Sector Pay

Figure 1 shows the global compensation distribution. It is bimodal-ish, with a primary mass between $70K and $100K and a secondary shoulder near $120K–$140K. The mean ($101,544) and median ($100,313) sit almost on top of each other, which is unusual for a salary distribution and tells us that the right tail, though it stretches to nearly $400K, contains too few employees to drag the mean meaningfully above the median. This is the empirical fingerprint of a capped grade-driven pay system, not a winner-take-all market.

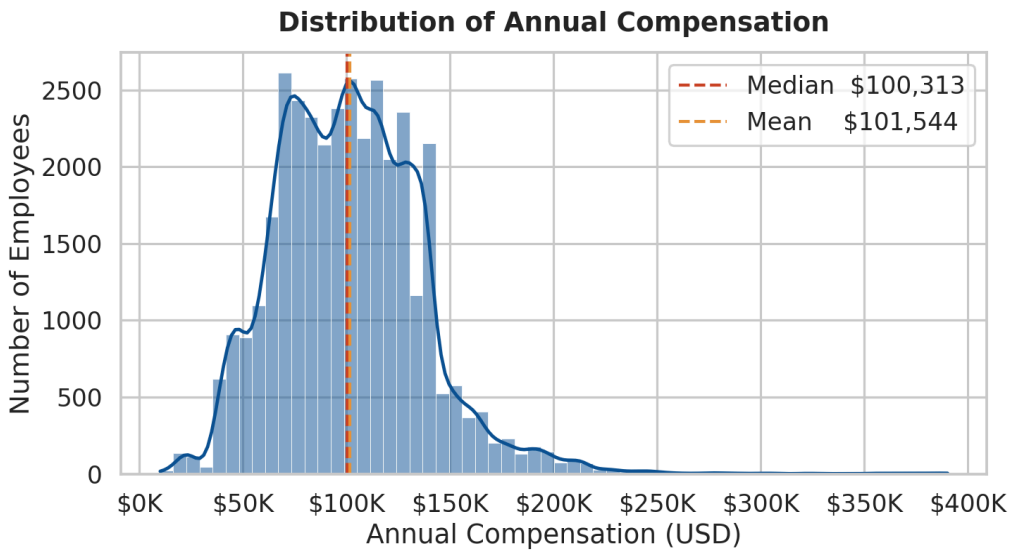

Figure 1. Distribution of annual compensation across the cleaned 35,627-employee dataset.

4.2 Pay by Appointment Category

Figure 2 stratifies pay by appointment category. The ordering is clean: Executive (Exe) appointments anchor the top of the distribution with medians above $200K, followed by Legal Service (LS), Management Supervisory Service (MSS), Excepted Service (ExS), Educational Service (Ed), and finally Career Service (CS). The Career Service category, despite having the lowest median, has the widest right tail with outliers reaching nearly $400K, almost certainly senior agency directors who entered the system before the Executive appointment type was widely used.

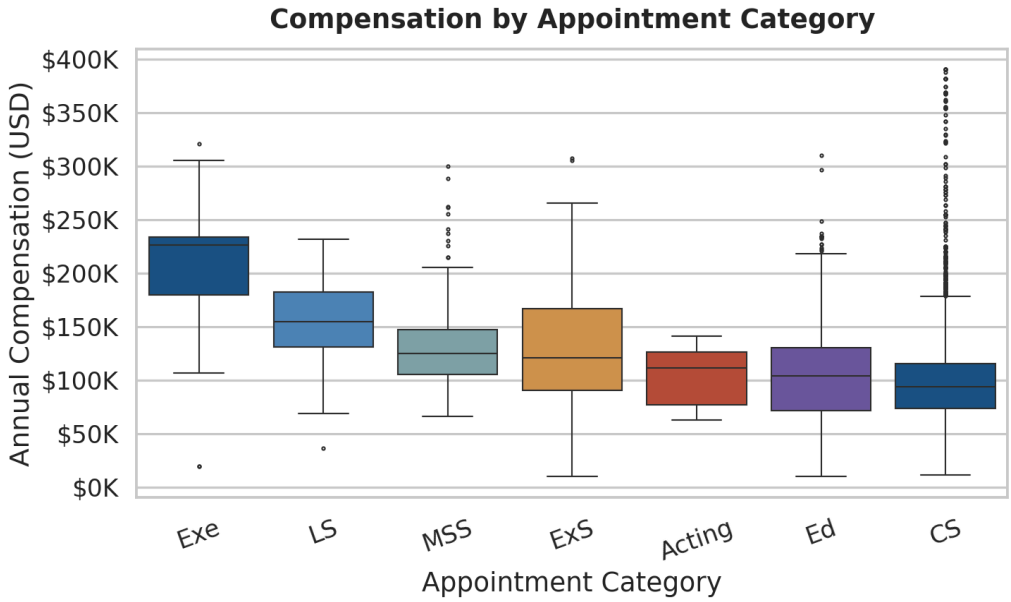

Figure 2. Compensation by appointment category, ordered by median.

4.3 Where Pay Concentrates: Agency × Appointment

Figure 3 cross-tabulates the two strongest grouping variables. Three patterns emerge. First, agency baseline matters: OCFO Career Service medians ($119K) exceed CS medians at almost every other agency, reflecting the financial-skill premium. Second, executive bands span a remarkable range across agencies — MPD's executive-tier pay tops $300K, more than DCPS's executive pay ($125K), suggesting differing executive structures by mission. Third, the matrix has structural empty cells; not every agency uses every appointment category, which is itself information about how each agency is organized.

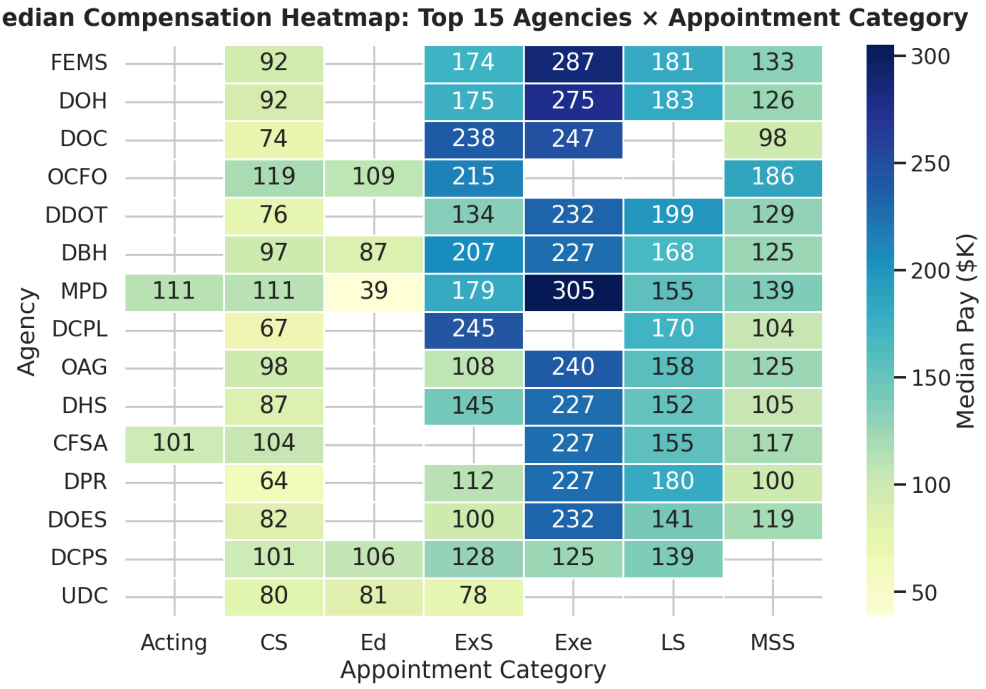

Figure 3. Median compensation by agency and appointment category (top 15 agencies).

4.4 Tenure Behaves Differently in Each Career Track

A pooled scatter of tenure against compensation yields a Pearson correlation of 0.293, modest, but real. That single number conceals very different career-arc shapes once we split by appointment category (Figure 4). Legal Service has by far the steepest slope: $125K at zero-to-two years, rising to over $200K past thirty years, a ~60% lifetime-pay gain. Career Service rises smoothly from $72K to a peak of $117K at twenty-to-thirty years, then dips slightly at thirty-plus, likely a survivorship effect. Management Supervisory Service is essentially flat from year one, it enters at ~$125K and stays there, indicating that MSS appointment is a level hire, not a ladder. Educational Service sits between CS and MSS.

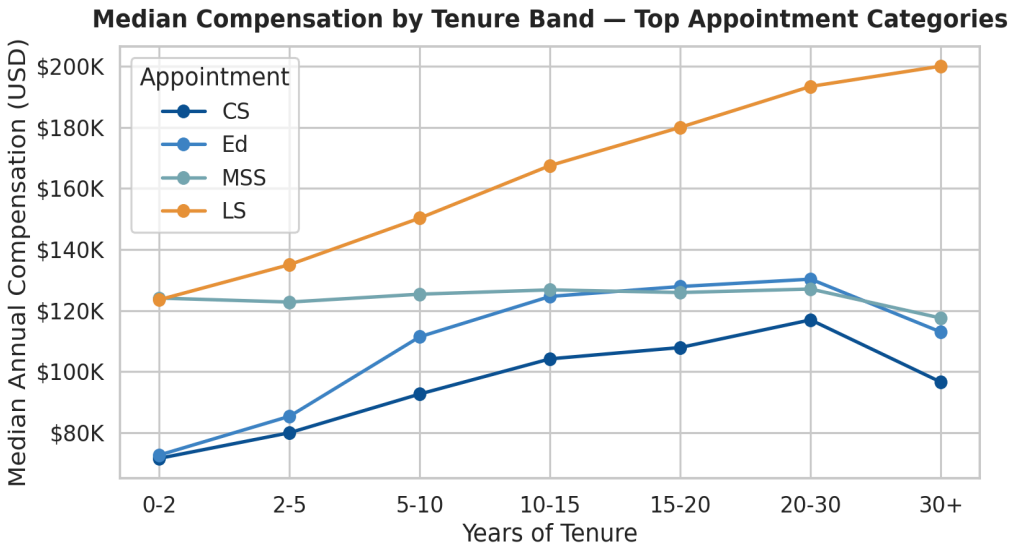

Figure 4. Median compensation by tenure band, separately for each top appointment category.

4.5 Internal Pay Dispersion

Figure 5 reports the P90/P10 ratio, the ratio of the ninetieth-percentile pay to the tenth-percentile pay, for the top fifteen agencies. DDOT (the District Department of Transportation) is a striking outlier at 5.8×, almost twice the median ratio of 2.1×. DDOT mixes professional engineers and senior planners with a large field crew of laborers, and the ratio captures that two-tier structure. DCPS (3.0×) and OCFO (2.7×) follow. By contrast, DOC (Department of Corrections) and CFSA (Child and Family Services) are the most internally compressed agencies at 1.7×, both are organizations dominated by a single skilled-but-not-executive role profile.

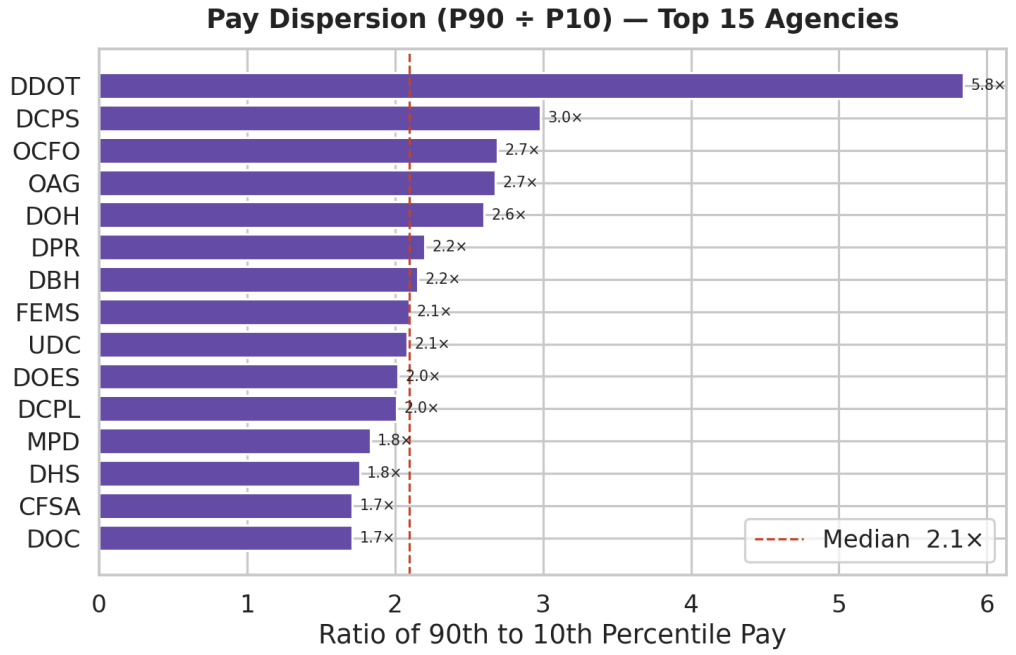

Figure 5. Pay dispersion (P90 ÷ P10) for the top fifteen agencies.

5. Conclusion

The headline result is that DC's public workforce is more compressed and more education-dominated than a casual look would suggest. DC Public Schools alone accounts for 28% of the cleaned dataset; together with the Metropolitan Police Department and Fire & EMS, three agencies absorb nearly half of all public employees in the District. The compensation distribution is near-symmetric around $100K with a contained right tail, the empirical signature of a strong grade-cap regime. Appointment category is the dominant axis of pay variation; agency baseline is a secondary but real effect.

The tenure-pay relationship is real but heterogeneous. Pooled, it produces a modest correlation of 0.293. Disaggregated by appointment category, it splits into very different shapes, a steep Legal Service ladder, a flat Management Supervisory plateau, and a slow-and-steady Career Service climb that peaks at the twenty-to-thirty-year mark. Pay dispersion within agencies ranges from 1.7× (DOC, CFSA) to 5.8× (DDOT), and the agencies with the highest dispersion are those that mix highly-credentialed professionals with large operational workforces.

Three caveats limit the analysis. The dataset is a single point-in-time snapshot, so individual pay trajectories cannot be inferred. The COMPRATE field is an annual rate, so part-time positions may be imperfectly normalized. And the dataset contains no demographic columns, race, gender, age, so well-documented questions about pay equity along those dimensions cannot be answered with this data alone. Future work pairing successive annual snapshots could turn this cross-section into the longitudinal panel that those questions demand.

6. References

DC Open Data — DC Public Employee Salary Dataset.
https://opendata.dc.gov/datasets/dc-employee-salary-information

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